

The Sequence-Quotient Reflection Theorem Fails for Compact Operators

Run fa_banach_001, lane 19

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Source Problem and Internal Lemma

The source paper is Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257. After proving the commutative and von Neumann boundary cases, it asks for a noncommutative C^* -algebra classification:

$\{U_\alpha\}_{\alpha \in A}$ of K is called a π -basis (weak basis) for K if every nonempty open subset of K contains some U_α in $\mathcal{U}$ [37, Page 15].

In the setting of commutative C^* -algebra, [25, Theorem 2.1] can be reformulated as follows. Note that [25, Theorem 2.1] only deals with real continuous functions. The complex case is a direct consequence of the real case.

Corollary 3.9. *Let $\mathcal{A}$ be a unital commutative C^* -algebra. Then the followings are equivalent:*

- (1) $\mathcal{A}$ has the BCP;
- (2) the pure state space $\mathfrak{P}(\mathcal{A})$ has a countable π -basis;
- (3) $\mathcal{A}$ has the UBCP.

In particular, there exist commutative C^* -algebras containing no minimal projections but having the BCP. For example, the uniform norm closure of the linear span of Haar functions [24, p.150].

We conclude this section by the following question.

Question 3. *It will be interesting to find a non-commutative version of Corollary 3.9, that is, how to characterize those C^* -algebras $\mathcal{A}$ having BCP/UBCP.*

This packet is not a counterexample to the source problem. It is a counterexample to an intermediate reflection lemma proposed during the lane-19 push:

If a countable BCP witness $C \subset A$ is captured by a unit vector in $\ell_\infty(A)/c_0(A)$, then C is captured by a unit vector in A .

Statement

Let $H = \ell_2$, let $A = \mathcal{K}(H)$, and let

$$A_\infty = \ell_\infty(A)/c_0(A).$$

Let $\Delta : A \rightarrow A_\infty$ denote the diagonal embedding.

Theorem 1. *There is a countable BCP witness $C \subset S_A$ and an element $v \in S_{A_\infty}$ such that*

$$\text{dist}(\Delta(c), \mathbb{C}v) = \|c\| \quad (c \in C),$$

but there is no $u \in S_A$ such that

$$\text{dist}(c, \mathbb{C}u) = \|c\| \quad (c \in C).$$

Idea of the Proof

Let p_n be the rank-one projection onto the n -th basis vector of ℓ_2 . Every compact operator is asymptotically disjoint from these rank-one projections:

$$\|a - \lambda p_n\| \longrightarrow \max\{\|a\|, |\lambda|\}.$$

Thus $v = [(p_n)]$ captures the diagonal copy of every compact operator in the sequence quotient.

On the other hand, if C is dense in the unit sphere of A , then no actual unit compact operator u can capture all of C , because some $c \in C$ is close to u , giving

$$\text{dist}(c, \mathbb{C}u) < \|c\|.$$

The Compact-Operator Limit

Lemma 1. *Let (e_n) be the standard orthonormal basis of ℓ_2 , and let p_n be the rank-one projection onto $\mathbb{C}e_n$. For every $a \in \mathcal{K}(\ell_2)$ and every $\lambda \in \mathbb{C}$,*

$$\lim_{n \rightarrow \infty} \|a - \lambda p_n\| = \max\{\|a\|, |\lambda|\}.$$

Proof. Let P_N be the projection onto $\text{span}\{e_1, \dots, e_N\}$. Since a is compact,

$$\|a - P_N a P_N\| \longrightarrow 0.$$

Fix $\varepsilon > 0$, and choose N with

$$\|a - P_N a P_N\| < \varepsilon.$$

For $n > N$, the operators $P_N a P_N$ and p_n act on orthogonal summands, so

$$\|P_N a P_N - \lambda p_n\| = \max\{\|P_N a P_N\|, |\lambda|\}.$$

Therefore, for $n > N$,

$$\begin{aligned} \left| \|a - \lambda p_n\| - \max\{\|P_N a P_N\|, |\lambda|\} \right| &\leq \|a - P_N a P_N\| \\ &< \varepsilon. \end{aligned}$$

Also $|\|P_N a P_N\| - \|a\|| < \varepsilon$ after increasing N , since $P_N a P_N \rightarrow a$ in norm. Letting $\varepsilon \downarrow 0$ proves the claim. $\square$

Proof of the Counterexample

Proof of Theorem 1. Let $C \subset S_A$ be a countable dense subset of the unit sphere of $A = \mathcal{K}(\ell_2)$. This is a BCP witness: for every $u \in S_A$, choose $c \in C$ with $\|c - u\| < 1/2$. Then

$$\text{dist}(c, \mathbb{C}u) \leq \|c - u\| < 1/2 < 1 = \|c\|.$$

Let $v \in A_\infty$ be the class of the bounded sequence (p_n) . Since $\|p_n\| = 1$, we have $v \in S_{A_\infty}$. For any $a \in A$ and $\lambda \in \mathbb{C}$, Lemma 1 gives

$$\|\Delta(a) - \lambda v\| = \limsup_n \|a - \lambda p_n\| = \max\{\|a\|, |\lambda|\}.$$

Taking the infimum over $\lambda \in \mathbb{C}$, we obtain

$$\text{dist}(\Delta(a), \mathbb{C}v) = \|a\|.$$

In particular this holds for every $c \in C$.

It remains to show that no actual unit vector in A reflects this capture. Let $u \in S_A$. Since C is dense in S_A , choose $c \in C$ with

$$\|c - u\| < 1/2.$$

Then

$$\text{dist}(c, \mathbb{C}u) \leq \|c - u\| < 1/2 < 1 = \|c\|.$$

Thus u does not capture C . Since $u \in S_A$ was arbitrary, no such reflecting unit vector exists in A . $\square$

Consequence for the C*-BCP Program

This counterexample shows that diagonal capture in $\ell_\infty(A)/c_0(A)$ is too weak to imply actual capture in A , even for a countable BCP witness. Therefore the proposed reflection shortcut cannot prove

$$\text{C*-algebra BCP} \Rightarrow \text{C*-algebra UBCP}.$$

There is no contradiction with known positive results: $A = \mathcal{K}(\ell_2)$ is separable, hence has UBCP. The point is subtler. UBCP spaces can still have sequence-quotient elements that capture countable diagonal sets. Therefore a proof of $\text{BCP} \Rightarrow \text{UBCP}$ must use more than the mere existence of sequence-quotient capture obtained by diagonalization.

Novelty and Search Bounds

Local indexes were searched for arXiv:2311.14257, reflection theorem, sequence quotient, countable capture, compact operators, BCP, and UBCP. The previous lane-19 attempts proposed the reflection route but did not record this compact-operator counterexample. This packet is a counterexample to an agent-generated intermediate lemma, not a new answer to the source paper's Question 3.

Human Review Focus

Reviewers should check:

1. the compact approximation step $P_N a P_N \rightarrow a$ for $a \in \mathcal{K}(\ell_2)$;
2. the orthogonal-block norm formula for $P_N a P_N - \lambda p_n$ when $n > N$;
3. the computation of distance to the line $\mathbb{C}v$ in the quotient;
4. the fact that a dense subset of the unit sphere is a BCP witness but cannot be actually captured by any unit vector.

References

- [1] Jinghao Huang, Karimbergen Kudaybergenov, and Rui Liu, *The ball-covering property of von Neumann algebras and noncommutative symmetric spaces*, arXiv:2311.14257, 2023.
- [2] John B. Conway, *A Course in Functional Analysis*, Springer, 1990.
- [3] Run `fa_banach_001`, lane 19, *BCP and Countable Capture Are Exact Opposites*, solution packet `2311.14257_bcp_countable_capture_exact_equivalence`, 2026.
- [4] Run `fa_banach_001`, lane 19, *Local Transition Bumps and the Reflection Wall*, attempt note `2311.14257_local_transition_bump_reflection_push`, 2026.